



The
University of
Faisalabad

IP Security

Lecture 22



Imagine you have different locks for different doors:

- **Email Lock (S/MIME, PGP)** → Only protects emails.
- **Login Lock (Kerberos)** → Only protects passwords.
- **Web Lock (SSL/HTTPS)** → Only protects websites.

Problem:

- If one lock fails, hackers can sneak in.
- Managing all these separate locks is complicated.
- Gaps between them can still leave you unsafe.

A Better Solution: Built-In Network Security

- Instead of adding separate locks to every door, **build security directly into the network itself**—like a **force field** that protects everything automatically.

How It Works:

- **All apps (email, web, logins) are secured by default.**
- No extra setup needed—just turn it on once.
- No weak spots between different tools.

Real-Life Example:

- Think of it like a **secure tunnel (like a VPN)** that all your internet traffic passes through. Instead of worrying about each app's security, the tunnel keeps everything safe—emails, logins, browsing—all at once.

Why This Is Better:

- **Stronger protection** (no gaps between tools).
Easier to manage (one system instead of many).
Works for everything (no need to secure each app separately).

This is called "**network-level security**"—making the internet itself safer, not just individual apps.

IP Security is like a secure bodyguard for your internet traffic. It protects data traveling between devices (like computers, phones, or servers) by adding three key features:

Authentication → *"Are you who you say you are?"*

- Checks if the sender/receiver is trusted (like a password for devices).

Confidentiality → *"No one can spy on your data."*

- Encrypts the data so hackers can't read it (like a secret code).

Key Management → *"Safe handshake for secure keys."*

- Creates and shares encryption keys securely (like a digital lock-and-key system).

Where Does IPSec Work?

- **LANs (Local Networks):** Offices, schools.
- **WANs (Wide Networks):** Private company networks or public internet.
- **The Internet:** Secures connections worldwide (e.g., remote work, VPNs).

Advantage of IP Sec

- **Strong Security for All Traffic**
 - Protects *every* data packet crossing the network (LAN/WAN/Internet).
- **Bypass-Resistant**
 - Hackers can't easily sneak past it (unlike app-specific security).
- **Works Automatically**
 - Runs below apps (no software changes needed).
 - Users don't even notice it (no extra steps).
- **User-Level Protection**
 - Can secure traffic for *specific people*, not just devices.
- **Safe Routing**
 - Guards against fake routes or hijacked data paths.

- **What is it?**

IPSec is like a **bodyguard for your internet connection** – it keeps your online traffic safe and private.

- **Why so complicated?**

- The rules are spread across **many technical manuals** (called RFCs).
- Some key ones: **RFC 2401, 2402, 2406, 2408** (but there are more).

How it protects you:

- **AH (Authentication Header)** → *Checks if data is real and hasn't been messed with.*
 - Like a seal on a package – if broken, you know it's been tampered with.
- **ESP (Encapsulating Security Payload)** → *Hides your data so no one can spy on it.*
 - Like sending a letter in a locked box instead of a postcard.

Where is it used?

- **IPv6 (new internet):** Must have IPSec (it's built-in).
- **IPv4 (old internet):** Can add IPSec if needed.

Best part?

- Works automatically – no extra apps or settings needed.
- Protects *everything* passing through (emails, web browsing, etc.).

1. Tunnel vs. Transport Mode

- IPsec has two ways to protect data. *Transport Mode* secures only the actual message (like locking a letter), making it faster for direct connections (e.g., office networks). *Tunnel Mode* protects everything—the message, sender, and receiver (like a locked delivery truck)—ideal for VPNs or public internet use.

2. Security Tools (AH & ESP)

IPSec uses two main tools. *AH* (**Authentication Header**) acts like a tamper-proof seal, ensuring no one altered the data. *ESP* (**Encapsulating Security Payload**) goes further: it encrypts the data (locks it) *and* checks for tampering, making it the stronger choice.

3. Security Databases (SPD & SAD)

IPSec relies on two "rulebooks." The *SPD* (**Security Policy Database**) decides *which traffic* gets protected (e.g., "secure all emails"). The *SAD* (**SA Database**) stores *how* to protect it (e.g., "use AES-256 encryption"). Together, they automate security without user effort.

4. Security Association (SA) & SPI

An SA is a secure agreement between devices, like a handshake saying, "Use these encryption keys." The *SPI* (**Security Parameter Index**) is a unique tag for each SA, helping devices quickly find the right rules for each connection.

5. NAT Issues (And Fix)

- NAT (what lets your whole home use one internet connection) messes up IPSec's AH security. The easy solution? **Just use ESP in Tunnel Mode instead!**
- Why?
- ESP hides *all* your data (like a secret tunnel)
- Works perfectly with NAT
- No complicated setup - just works!
- Best for:
 - ✓ Home WiFi
 - ✓ Office networks
 - ✓ VPN connections

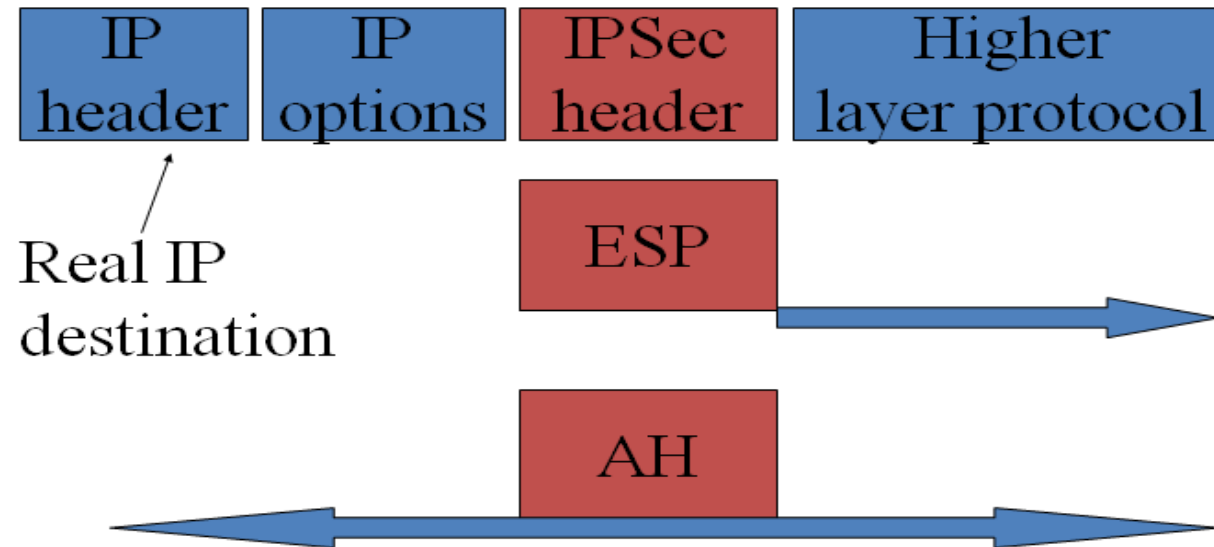
Transport Mode (Like a locked envelope)

- Only protects the actual message inside
- Faster but less secure
- Good for: Office networks where devices trust each other

Tunnel Mode (Like a locked delivery truck)

- Protects everything: message + sender/receiver info
- Slower but super secure
- Good for: VPNs, public WiFi, internet connections
- Key difference:
Transport Mode = Just protect the letter
Tunnel Mode = Protect the whole package

Transport Mode



ESP (Encapsulating Security Payload)

Protects only the actual message content (higher layer data)

Provides both encryption (hides data) and integrity checking (detects tampering)

Does not protect the IP address information in packet headers

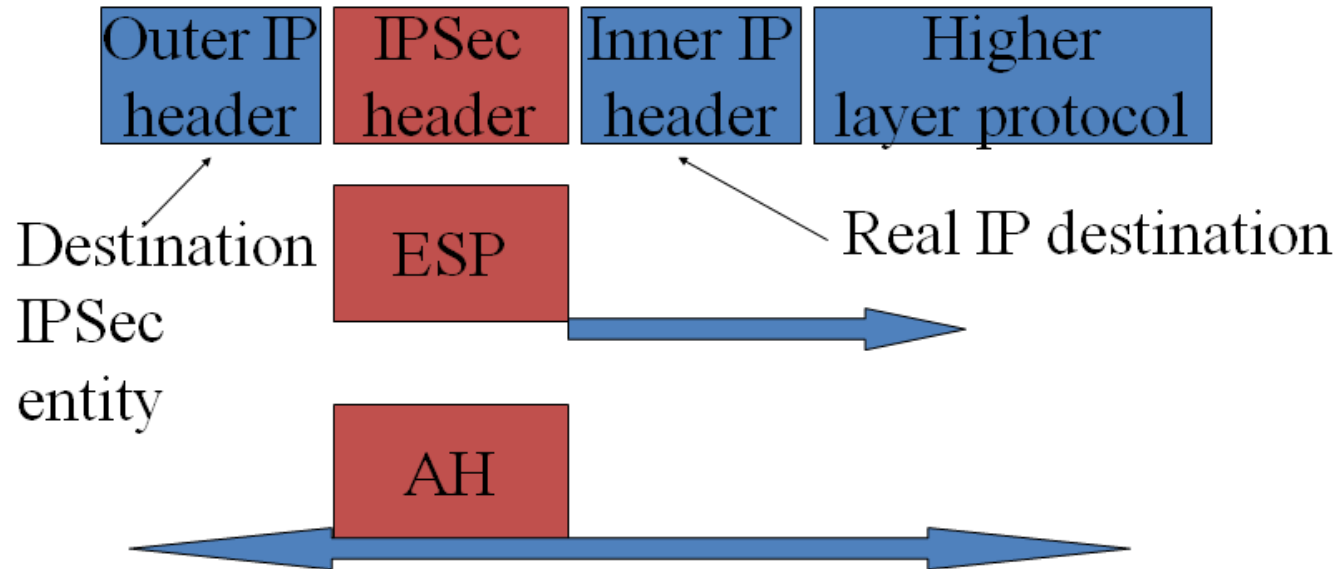
AH (Authentication Header)

Protects both the message content AND the IP header information

Provides integrity checking for the entire packet (detects tampering)

Does not provide encryption (data remains visible)

Tunnel Mode



- ESP applies only to the tunneled packet
- AH can be applied to portions of the outer header

Security Association (SA)

- A **Security Association (SA)** defines how IPSec protects traffic between devices using three key parameters: the **Security Parameters Index (SPI)** to identify the SA, the **destination IP address**, and the **security protocol** (AH or ESP). SAs are stored in a database and determine how senders encrypt data and how receivers decrypt it. Unlike fixed rules, SAs are dynamically created and tailored for each traffic flow, ensuring customized security for different connections.

- The **Security Parameters Index (SPI)** is a 32-bit identifier that helps the receiver pick the right **Security Association (SA)** to process incoming IPSec packets. Sent with each packet, the **SPI—combined with the destination IP and IPSec protocol (AH/ESP)—uniquely identifies an SA**, ensuring secure communication as agreed between sender and receiver.

SA Database (SAD)

- The **SA Database (SAD)** stores the security parameters for each **Security Association (SA)**, including:
 - **Lifetime** of the SA
 - **AH/ESP encryption & authentication details**
 - **Mode** (Tunnel or Transport)
- Every IPSec-enabled device (hosts/gateways) maintains its own SAD to manage active secure connections.

- The **Security Policy Database (SPD)** determines which traffic gets IPSec protection and how. It contains rules that specify:
- **What data to secure** (based on IPs, ports, protocols)
- **Which Security Association (SA) or SA bundle to apply**
Each device maintains its own SPD, using **selector fields** like source/destination IP, transport protocol, and ports to match traffic to the right security rules.

- *SAs are flexible IPSec "contracts" between devices, using SPI + IP + protocol to secure traffic uniquely for each flow.*
- *SPI is a unique tag that, along with IP and protocol, tells the receiver which SA to use for decryption.*
- *SAD is a device's personal "rulebook" for each secure connection, tracking how and how long to protect data*
- *SPD is a "traffic filter" that decides which connections get IPSec protection and how.*

Question?